TECHNISCHE UNIVERSITÄT BERLIN

Fakultät VI

Assignment II

„ Intelligent Transport Systems “
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eingereicht bei

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1. Shortest path to Cottbus

Table 1. Label-correcting iterations for the shortest-path calculation

Iteration	Remove	OPEN	dS	d1	d2	d3	d4	d5	d6	d7	d8	d9	d10	dT
1	/	S	0	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞
2	S	1,2,3	0	7	3	1	∞	∞	∞	∞	∞	∞	∞	∞
3	1	2,3,4,5	0	7	3	1	11	9	∞	∞	∞	∞	∞	∞
4	2	1,3,4,5	0	5	3	1	11	9	∞	∞	∞	∞	∞	∞
5	1	3,4,5	0	5	3	1	9	7	∞	∞	∞	∞	∞	∞
6	3	4.5	0	5	3	1	9	4	∞	∞	∞	∞	∞	∞
7	4	5,6,7	0	5	3	1	9	4	15	11	∞	∞	∞	∞
8	5	6,7,8	0	5	3	1	9	4	15	11	15	∞	∞	∞
9	6	7,8,9	0	5	3	1	9	4	15	11	15	16	∞	∞
10	7	8.9	0	5	3	1	9	4	15	11	15	14	∞	∞
11	8	9,10	0	5	3	1	9	4	15	11	15	14	17	∞
12	9	10	0	5	3	1	9	4	15	11	15	14	17	19
13	10	-	0	5	3	1	9	4	15	11	15	14	17	19

Based on the label correcting algorithm (Table 1) the shortest path is:

$$S \to 2 \to 1 \to 4 \to 7 \to 9 \to T$$

The minimum cost of travel is 19.

2. Selling the ticket

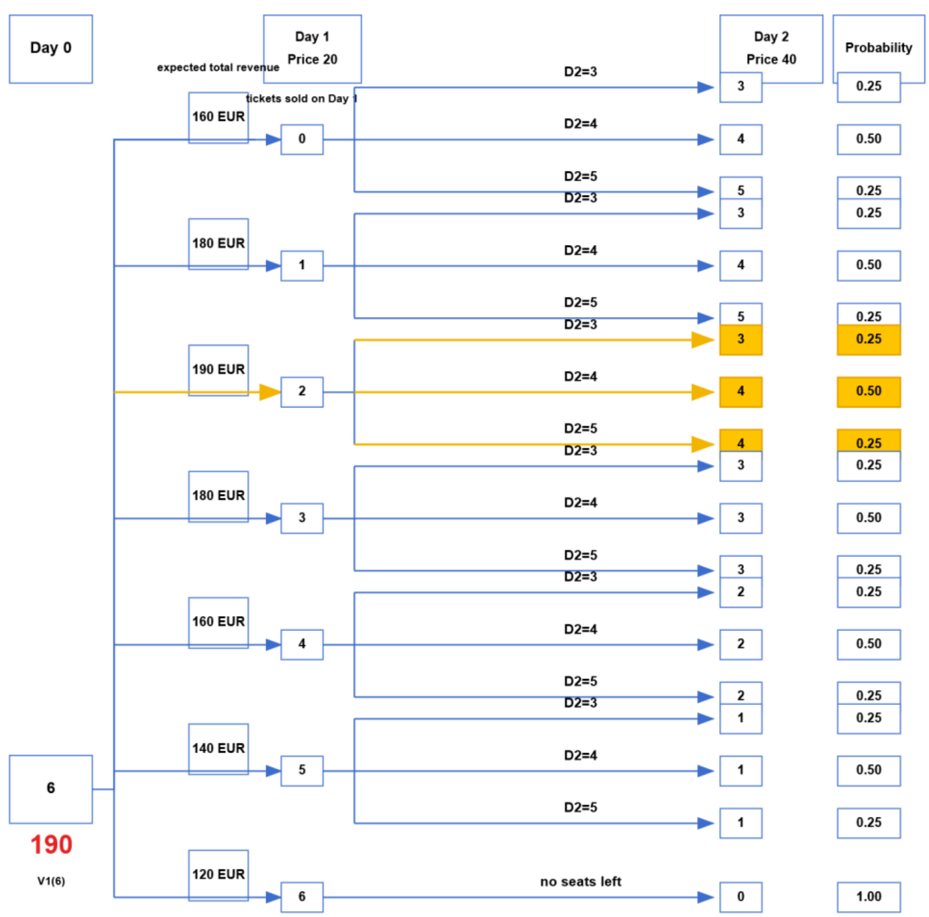

Figure 1. Dynamic-programming evaluation of day-1 ticket sales and expected revenue

As shown in Figure 1, the optimal number of tickets to put up for sale on day 1 is: 2 tickets

$$E[min(4, D2)] = 0.25 \cdot 3 + 0.5 \cdot 4 + 0.25 \cdot 4 = 3.75$$

$$40 + 150 = 190$$

Expected revenue is 190 Euro.

3. The talk on DRT service in Cottbus

3.1 Preliminary Study

The median passenger waiting time in the preliminary run was 466 s, or 7.77 min.

The preliminary run served all 10,000 trip requests without any rejections, which indicates that the system had sufficient capacity. The fleet travelled about 38,134 km in total, including roughly 2,041 km without passengers. The empty-distance share was therefore only about 5%. Most vehicle kilometres were associated with passenger service, suggesting that the dispatching process used the fleet efficiently. The passenger-kilometre-to-vehicle-kilometre ratio was 2.23, which also shows a clear pooling effect and suggests that shared DRT trips can reduce some of the traffic generated by individual travel.

The system nevertheless used small 300 vehicles and produced a substantial amount of vehicle travel. If DRT replaced conventional buses completely, the road network could carry more small vehicles instead of a smaller number of high-capacity buses. DRT may therefore improve service flexibility while increasing traffic volumes, particularly during peak periods, around pick-up and drop-off locations, and in areas where demand is concentrated. These conditions could cause local congestion and kerbside disturbance.

Based on the preliminary run, an eight-seat minivan or small shuttle is recommended. The baseline fleet of 300 eight-seat vehicles served all 10,000 requests with no rejections, a median waiting time of about 7.8 min, and an empty-distance share of about 5%. An eight-seat vehicle offers more scope for pooling than a standard taxi, while remaining more flexible than a bus or a large shuttle for door-to-door operation. It therefore provides a reasonable balance between service quality, pooling capacity, and operational flexibility.

3.2 Systematic Study

3.2.1 Improving Service Quality

(1) Evaluation Criteria

Following the evaluation framework used in recent research on on-demand public transport [1], six criteria were selected to cover passenger experience, reliability, operational efficiency, and transport efficiency: average waiting time, P95 waiting time, average total travel time, number of rejected requests, minimum idle vehicle share, and the passenger-kilometre-to-vehicle-kilometre ratio (d_p/d_t). The first five criteria were treated as cost criteria, while d_p/d_t was treated as a benefit criterion. Since the city had not specified a policy preference among these dimensions, all six criteria were given equal weight.

Table 2. Service quality evaluation criteria

Domain	Criteria	Weights
Passenger experience	Average waiting time	1/6
	P95 waiting time	1/6
	Average total travel time	1/6

Service reliability	Rejected requests	1/6
Operational efficiency	Minimum idle vehicle share	1/6
Transport efficiency	d_p/d_t	1/6

(2) Parameter Sets and Simulation Results

The systematic study varied three parameters: fleet size, vehicle capacity, and maximum waiting time. Together with the preliminary setting, these choices produced nine scenarios, S1-S9. Table 3 lists the parameter combinations and the main simulation results.

Table 3. Parameter settings and simulation results for S1-S9

Set	Fleet	Seats	Max wait time	Avg wait time	P95 wait time	Total travel time	Rejections	Minimum idle vehicle share	dp/dt
S1	300	8	1200s	8.6763	18.6667	22.8962	0	0.43	2.23
S2	300	6	1200s	8.4957	18.6325	22.7875	0	0.41	2.26
S3	300	4	1200s	9.0293	18.7833	22.5855	0	0.38	2.08
S4	250	8	1200s	8.9497	18.7833	23.1285	0	0.29	2.19
S5	250	6	1200s	8.8193	18.7333	22.9623	0	0.28	2.19
S6	250	4	1200s	9.2223	18.7667	22.8735	0	0.27	2.07
S7	300	8	900s	7.5105	14.5333	22.3413	1	0.41	2.36
S8	300	8	700s	6.5393	11.3833	21.8253	5	0.41	2.43
S9	250	6	900s	7.788	14.5333	22.6343	1	0.29	2.31

Note: Blue cells mark the maximum-wait comparison; orange cells mark the seat-capacity comparison with 300 vehicles; purple cells mark the fleet-size comparison; green cells mark the seat-capacity and operating-result comparison with 250 vehicles.

(3) AHP Evaluation of the Parameter Sets

The Analytic Hierarchy Process (AHP) was used to compare the scenarios across the six criteria. Pairwise preferences were expressed on Saaty's 1-9 scale [2]. A value of 1 means that two alternatives are equally preferred, while 9 indicates an extreme preference for one alternative. Even numbers represent intermediate judgements. The scale is summarized in Table 4.

Table 4. Saaty's 1-9 pairwise comparison scale

Intensity	Definition	Explanation
1	Equal importance	Two activities contribute equally to the objective
3	Moderate importance	Experience and judgment slightly favor one over another
5	Strong importance	Experience and judgment strongly favor one over another
7	Very strong importance	Activity is strongly favored, and its dominance is demonstrated in practice
9	Absolute importance	The importance of one over another is affirmed in the highest possible order
2, 4, 6, 8	Intermediate values	Used to represent a compromise between the priorities listed above

For each criterion, the scenarios were first ordered according to whether a lower or higher value was preferred. Their relative differences were then mapped to the 1-9 scale. If scenario i has a preference value a_{ij} over scenario j, the reverse comparison is $a_{ji} = 1/a_{ij}$, and each diagonal element is $a_{ii} = 1$. The same relative-gap thresholds were used for all criteria.

$$\Delta_{ij} = |v_i - v_j| / \min(|v_i|, |v_j|)$$
$$a_{ji} = 1/a_{ij},\ a_{ii} = 1$$

Applying the reciprocal condition gives the following matrix:

```
# Average waiting time (cost criterion): S1-S9 pairwise comparison matrix
avg_wait <- matrix(c(
  1, 0.5, 3, 3, 2, 3, 0.2, 0.1429, 0.25,
  2, 1, 3, 3, 3, 4, 0.2, 0.1667, 0.25,
  0.3333, 0.3333, 1, 1, 0.5, 2, 0.1667, 0.1429, 0.2,
  0.3333, 0.3333, 1, 1, 0.5, 3, 0.2, 0.1429, 0.2,
  0.5, 0.3333, 2, 2, 1, 3, 0.2, 0.1429, 0.2,
  0.3333, 0.25, 0.5, 0.3333, 0.3333, 1, 0.1667, 0.1429, 0.2,
  5, 5, 6, 5, 5, 6, 1, 0.2, 3,
  7, 6, 7, 7, 7, 7, 5, 1, 5,
  4, 4, 5, 5, 5, 5, 0.3333, 0.2, 1
), nrow = 9, byrow = TRUE,
dimnames = list(alternatives, alternatives))
APCL[["avg_wait"]] <- avg_wait
```

Because the six criteria were assigned equal importance, their pairwise comparison matrix is:

```
# Equal criterion weights: all six criteria are treated as equally important
criteria <- names(APCL)
CWPC <- matrix(
  1,
  nrow = length(criteria),
  ncol = length(criteria),
  dimnames = list(criteria, criteria)
)
criteria_weights <- getWeights(CWPC)
```

Figure 2 shows the final scores. S8 received the highest score (23.4%), followed by S7 (13.4%) and S9 (12.6%).

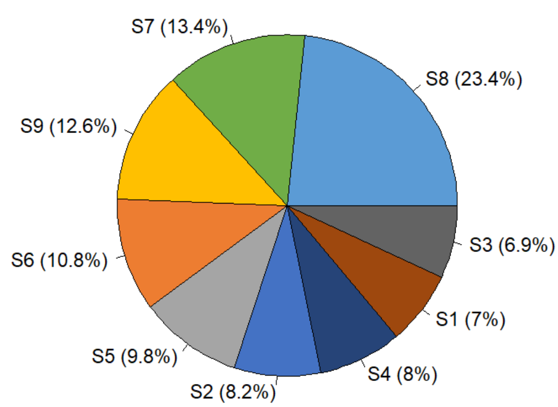

Figure 2. AHP ranking of S1-S9 DRT scenarios

(4) Findings

The coloured cells in Table 3 indicate the four comparisons discussed below.

a. The blue cells compare S1, S7, and S8. With fleet size and vehicle capacity fixed, reducing the maximum waiting time from 1,200 s to 900 s and 700 s decreased the average waiting time from 8.6763 min to 7.5105 min and 6.5393 min, while P95 waiting time fell from 18.6667 min to 14.5333 min and

11.3833 min. However, rejected requests increased from 0 to 1 and 5. Thus, tighter waiting-time limits improve service for accepted passengers but make full request acceptance harder.

b. The orange cells show that larger vehicles are not always better, although four seats appear to limit pooling. Among the 300-vehicle scenarios, S2 with six seats slightly outperformed the eight-seat baseline S1 in average waiting time (8.4957 vs. 8.6763 min) and d_p/d_t ratio (2.26 vs. 2.23). In contrast, S3 with four seats had a longer average waiting time of 9.0293 min and a lower d_p/d_t ratio of 2.08. At this demand level, six to eight seats seem sufficient, while four seats reduce matching flexibility.

c. The purple cells compare S1 and S4. Both used eight-seat vehicles and a 1,200 s waiting-time limit, but S4 reduced the fleet from 300 to 250 vehicles. The minimum idle share decreased from 0.43 to 0.29, indicating higher utilisation. However, average waiting time rose from 8.6763 min to 8.9497 min, and average total travel time from 22.8962 min to 23.1285 min. Therefore, improved vehicle utilisation came with a small reduction in passenger service quality.

d. The green cells compare S4-S6 with 250 vehicles. S5, with six-seat vehicles, provided the most balanced result: it had the lowest average waiting time among the three scenarios (8.8193 min), zero rejections, a minimum idle share of 0.28, and $d_p/d_t = 2.19$. If the aim is to reduce fleet size without rejecting requests, 250 six-seat vehicles are a plausible lower-cost option.

Overall, no scenario performed best on all criteria. Shorter waiting-time limits improved accepted passengers' experience but increased rejection risk. A smaller fleet reduced idle share but increased waiting and travel times. The final parameter choice therefore depends on the city's priorities regarding zero rejections, waiting time, and fleet cost.

3.2.2 Pricing Scheme

(1) Cost Estimation

Based on the DRT cost analysis method proposed by McCollom and Polin (1988), the annual total cost can be estimated using three main variables: operating hours, operating mileage, and the number of vehicles in service [3].

$$\begin{aligned} &Annual\ Total\ Cost \\ &= Cost\ per\ hour \times Annual\ hours\ of\ operation \\ &+ Cost\ per\ mile \times Annual\ miles\ of\ operation \\ &+ Cost\ per\ vehicle\ operated \times Number\ of\ vehicles\ in\ service \end{aligned}$$

In this study, the hourly cost is represented by driver labor cost, the mileage cost is represented by vehicle operating cost, and the fixed vehicle cost is represented by depreciation and insurance.

The German Federal Employment Agency's Entgeltatlas reports that the full-time median monthly gross salary for Busfahrer/in is 3,522 EUR [4]. Germany Trade & Invest states that the employer's social insurance contribution in Germany is approximately 21% of the employee's gross wage [5].

Therefore, c_h is set to 25 EUR/vehicle-hour. We assume that the service operates from 05:00 to 23:00, i.e., 16 operating hours per day.

For vehicle cost, S8 uses 8-seat vehicles. Therefore, the Mercedes-Benz Vito Tourer kompakt 116 CDI Pro 9G-TRONIC is selected as a representative vehicle type. According to the ADAC cost table, the total vehicle cost for this model is approximately 91.7 ct/km [6]. The fixed cost per vehicle is estimated using a depreciation cost of 646 EUR/month and an insurance cost of 195 EUR/month.

The DREAM_PACE report indicates that digital DRT systems require a travel dispatch center, in-vehicle devices, user interfaces, and communication networks. Therefore, an additional 10% is added for dispatching and back-office management costs [7].

Table 5. Cost assumptions for the S8 DRT scenario

Parameter	Assumption	Basis
c_h	25 €/vehicle-hour	driver wage + employer contribution
c_{km}	0.917 €/vehicle-km	ADAC Vito Tourer total car cost
c_v	841 €/vehicle-month	depreciation + insurance
Overhead	× 1.10	dispatching, customer service, administration

Based on the assumptions in Table 5, the estimated annual operating cost of the S8 scenario is 65.09 million EUR/year.

In S8, the model reports 9,995 served DRT rides per day and 5 rejected requests. Therefore, the cost-recovery fare is approximately 17.8 EUR/ride.

(2) Fare Comparison

Cottbus public transport fares are 2.70 EUR for a single ticket, 5.60 EUR for a 24-hour ticket, and 39.90 EUR for a monthly ticket [8].

The daytime taxi tariff includes a base fare of 3.40 EUR, 2.60 EUR/km for the first 2 km, and 2.00 EUR/km after 2 km. Using an average passenger trip distance of 3.6 km, the estimated taxi fare is approximately 11.80 EUR [9].

To fully cover the estimated annual operating cost, the DRT fare would need to be set at approximately 18 EUR per ride.

Table 6. Fare comparison between DRT and existing transport options

Mode / fare type	Fare level
Cottbus public transport single ticket	2.70 €
Cottbus 24-hour ticket	5.60 €
Average 3.6 km taxi trip	≈ 11.80 €
Full cost-recovery DRT fare	18 €

As shown in Table 6, the full cost-recovery fare for S8 is much higher than existing public transport fares and also higher than the estimated fare for an average 3.6 km taxi trip. Therefore, although a fare of about 18 EUR/ride could recover the full operating cost, it is not price competitive. Former bus users would likely consider it too expensive.

(3) Reasons for High Cost and Cost-Reduction Measures

Driver cost accounts for about 67% of total operating cost. Passengers pay not only for their own ride, but also for the availability of the whole DRT fleet.

To reduce operating costs, a dynamic fleet operation strategy could first be adopted. More vehicles could be deployed during peak periods, while the number of active vehicles could be reduced during off-peak periods, thereby reducing driver-hours. Second, the waiting-time target could be relaxed moderately, accepting a slightly longer passenger waiting time in exchange for a lower vehicle requirement. Third, a mixed fleet could be used, with cheaper 4-seat or 6-seat vehicles assigned to low-demand areas to reduce fixed vehicle costs. In the long term, autonomous DRT could also reduce operating costs significantly by removing driver cost, but due to regulatory, safety, and technical constraints, it is unlikely to be the main solution in the short term.

- Material: All scripts, outputs and Video are on Github [Yuuka-xi/ITS-Homework2: ITS-DRT Group Project]

References

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